**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 003 · Hexadecimal, shifts & masks

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Choose a type

Q2. Read an unsigned byte

Q3. Encode a negative number

Q4. Add and check the range

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

Recall the term — Hexadecimal

Recall the term — Nibble

Recall the term — Logical shift

Recall the term — Mask

Recall the term — AND

Recall the term — OR

Recall the term — XOR

## New learning

Hexadecimal links directly to binary — Each hexadecimal digit maps to exactly four bits because four bits have 16 possible patterns. — CODE:
$6D_{16}$
$6 \rightarrow 0110$
$D \rightarrow 1101$
$= 01101101_2$

Logical shifts move bits — A logical shift inserts zeros into the empty positions. Bits leaving the fixed width are lost.

AND and OR masks — AND: mask 1 preserves a bit; mask 0 clears it. — OR: mask 1 sets a bit; mask 0 preserves it.

XOR toggles selected bits — XOR produces 1 when the two input bits differ. A mask bit of 1 flips the corresponding data bit; 0 leaves it unchanged.

## Worked example

Link denary, hexadecimal and binary

1. Find the hex digits — CODE:
$173 \div 16 = 10$ remainder 13 — The quotient gives the first hex digit; the remainder gives the second.

2. Convert 10 and 13 to hex — CODE:
$10 = A$
$13 = D$

$173_{10} = AD_{16}$

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3. Use four bits per hex digit — CODE:
A → 1010
D → 1101

$AD_{16} = 10101101_2$

4. Return to denary — CODE:
(10 × 16) + 13 = 173

128 + 32 + 8 + 4 + 1 = 173 — For binary → hex, group bits from the right in fours and pad the left group with zeros if needed.

## We Do · Practise with guidance

2. Use hexadecimal place values

Use the method from I Do to build the answer.

Build the answer

Reverse check: 755 ÷ 16 gives remainders 3, F, 2.

3. Trace logical shifts in a fixed byte

The third shift discards the leftmost 1, so 352 cannot fit in 8 bits.

4. Preserve or clear selected bits with AND

The mask keeps the low nibble and clears the high nibble.

5. Set with OR; toggle with XOR

Divide using a shift

44 ÷ 4 = 11; the discarded bits are both zero.

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# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

## Question 1

Hexadecimal → binary
Write the binary equivalent of hexadecimal B7E .
Paraphrased exam question; original value and mark retained. OCR H446/01, page 17
.

## Question 2

Denary → hexadecimal
Express $188_{10}$ in hexadecimal.
Paraphrased exam question; original value and mark retained. OCR H446/01, page 12
.

## Question 3

Show how the bases connect
Convert $11100101_{2}$ to hexadecimal. Show the two nibble values. [2]
Convert $3A7_{16}$ to denary. Show the place-value calculation. [2]
Convert $214_{10}$ to hexadecimal. Show quotient and remainder or binary grouping. [2]
Original classroom extensions of the conversion skills in Questions 1–2.

## Question 4

Track the stored bits
Logically shift 00110100 two places right. Give the eight-bit result and its denary value. [2]
Logically shift 10100001 one place left. Give the stored byte and explain why its

denary value is not double the starting value. [3]
Give the stored byte after logically shifting 00001111 one place right. [1]
Adapted from OCR June 2017 Q6(c): values changed; left-shift and precision extensions added. Classroom marks.

### Question 5

Apply each operation independently
Start with 10100101 each time. Give the result of:
AND with 00001111 . [1]
OR with 01000010 . [1]
XOR with 11110000 . [1]
Original exam-style practice aligned to PMT bitwise manipulation and masks.

### Question 6

Control an eight-bit register
Bits are numbered 7…0 from left to right. Give an operation and an eight-bit mask for each independent task. Preserve all bits not mentioned.
Clear bits 7 and 6. [2]
Set bits 3 and 0. [2]
Toggle bits 5 and 2. [2]
Original exam-style application; one mark for the operation and one for the mask.

## Exit Ticket · Answer before checking

One hexadecimal digit represents four binary bits.

**BLACK PEN** Answer all questions first.

**GREEN PEN** Then: tick correct answers; rewrite mistakes.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

## Page 1 onwards · Do Now

Q1. Boolean: True or False.

Q2. 64 + 16 + 8 + 2 = 90 .

Q3. +18 = 00010010 . Sign magnitude: 10010010 . Two’s complement: invert → 11101101 , add 1 → 11101110 .

Q4. 10010000 . 112 + 32 = 144, outside −128…127. Signed overflow occurs; the stored pattern represents −112 when interpreted as an 8-bit two’s-complement integer.

## You Do pages · Independent questions

Q1. B = 1011; 7 = 0111; E = 1110. Answer: 1011 0111 1110 (1). Retain four bits per hexadecimal digit.

Q2. 188 ÷ 16 = 11 remainder 12 → BC (1). Check: 11 × 16 + 12 = 188.

Q3. 1110 = E and 0101 = 5 (1); E5 (1).
(3 × 256) + (10 × 16) + 7 (1) = 935 (1).
214 ÷ 16 = 13 remainder 6 (1); D6 (1).

Q4. 00001101 (1), 13 (1).
01000010 (1). The leftmost 1 is shifted out and discarded (1); 161 × 2 = 322 exceeds 255, while the stored value is 66 (1).
00000111 (1). The stored value is 7, not 7.5.

Q5. CODE:
10100101 10100101 10100101
AND 00001111 OR 01000010 XOR 11110000
---------- ---------- ----------
00000101 11100111 01010101
One mark per correct result. Each operation starts from the original byte.

Q6. AND (1) with 00111111 (1). Zeros clear bits 7 and 6; ones preserve the rest.
OR (1) with 00001001 (1). Ones set bits 3 and 0.
XOR (1) with 00100100 (1). Ones toggle bits 5 and 2.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Q1–Q2 retain the original exam mark allocations. Q3–Q6 use classroom allocations. Answers are calculated teaching guidance, not a reproduced official OCR mark scheme.

### Exit Ticket · Model answer

True. Each hexadecimal digit represents four binary bits.

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**